Canonical Certification Does Not Exhaust Reflective Structure: Reflective Split, Strict Refinement, and Fiber Structure in Infinity Compression

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Abstract

Formal theories often collapse many derivations to a single canonical certified record: the question is whether that *collapse* exhausts everything structurally significant about the realized routes that produced it. This paper is the *internal origin theorem* of a larger program: we prove that it does not, on named machine-checked carriers in the Infinity Compression native environment. We prove universal collapse of standard Phase-2 extraction to a unique bare certificate; a reflective split (canonical bare certification alongside nontrivial enriched autonomous mirror structure); strict refinement of the typed forgetful map π_A (existence of distinct enriched reflective splits with equal π_A-image); and a forgetful–fiber layer in which the canonical fiber is nontrivial and role assignments inject into a distinguished standard subfamily. We treat a concrete NEMS-style flagship spine and extend under explicit intermediate crown hypotheses (role-rich and role-pair-diverse classes). Methodologically, the proofs are organized by collapse audits and carrier-sensitive route comparison. The result is a formal theorem of separation between theorem-level invariants and realization-level structure. Companion papers show that the same collapse/refinement/fiber architecture transfers to external mathematics across twelve benchmark families and yields theorem-level results in group extensions, embedding problems, and Quillen's Theorem A for Galois connections. All main claims are machine-checked in Lean 4 (zero `sorry` in proof terms under `InfinityCompression/`; Mathlib and kernel axioms only); Section A maps paper statements to named definitions and theorems in the artifact.

Trust boundary: Theorem statements are tied to explicit formal interfaces and hypotheses; companion transfer papers test portability, not automatic universality over all mathematics.

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Suite context. The Infinity Compression paper suite is organized as follows: the *flagship manuscript* states the *internal origin theorem*; the *transferability paper* validates the same packaging on twelve external benchmarks; *general-method manuscripts* (group extensions; Quillen) supply theorem-level leverage as the *algebraic and arithmetic engine* and *topology discharge*; the *synthesis manuscript* maps the full program; the *summit manuscript* distills reflective non-exhaustion; the *technical Mathlib companion* is reviewer-facing. This manuscript is the flagship *internal origin theorem* paper.

Keywords: formal verification, proof architecture, certification vs. realization, fiber structure, forgetful maps, Lean 4, Mathlib

MSC 2020: 03B70, 18A30, 68V15, 68V20

Code/Data availability: Repository: https://github.com/novaspivack/infinity-compression-lean. Root module: `InfinityCompression.lean`. Lean toolchain: `leanprover/lean4:v4.29.0-rc6`. Zero `sorry` in proof terms under `InfinityCompression/`.

1 Premises, disagreement coordinates, and trust boundary

Where disagreement can (and cannot) live

Formal spine. Machine-checked claims refer to named carriers and hypotheses inside `InfinityCompression/`; crown-eligible and role hypotheses are part of the statement ecology, not hidden assumptions.

Premise coordinates (for external readers).

- **Architecture scope.** Results are conditioned on the Infinity Compression interfaces and the standard bundle used in the development. *Logical effect:* Declining one hypothesis class does not invalidate unrelated companion manuscripts.
- **Living artifact.** Mathlib revision and module paths are pinned in `lakefile.lean` and repository documentation. *Logical effect:* Use the code-availability URL and appendix index (Section A) for referee checks.

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2 Introduction

2.1 The problem

Reflective or verification-oriented theories often end with a *principal certified invariant*: a theorem-level closure statement packaged as a canonical bare record (proof-irrelevant at that carrier). Whether such certification *exhausts* structurally significant information is then a precise mathematical question. If all admissible routes to the same certified conclusion were indistinguishable at every informative layer, closure would flatten the theory; otherwise certification and realization must split in a controlled, provable way.

In the Infinity Compression formalization, standard-bundle extraction to the bare carrier collapses to a *unique* certificate, while a proof-relevant enriched layer retains structure above that collapse (Section 4–Section 7).

2.2 Main thesis

The paper's claim (aligned with the title) is a *provable split*: theorem-level canonical bare certification versus enriched reflective realization. The typed forgetful map $\pi_A : \mathcal{E}(A) \to \mathcal{B}_{\text{can}}(A)$ is sound but not injective on standard examples; fibers over the canonical bare basepoint are nontrivial, with injective role data along a named standard subfamily. Theorem 2.1 states the cluster; Section A gives Lean anchors.

2.3 Main results

Theorem 2.1 (Main results)**.** *Fix a compression architecture A and the standard Phase-2 / roles+extraction interfaces of the development. Let $\pi_A : \mathcal{E}(A) \to \mathcal{B}_{\text{can}}(A)$ be the forgetful map* (Section 3.2)*. The load-bearing spine is:*

(1) ***Collapse / canonical certification:*** *for all skeletons s and extractions H,* $\text{admissibleFromSkeleton}(s, H) = b_A^*$*; all bundled bridge derivations lie in the canonical bare class* Can (Theorem 4.1, Theorem 4.3)*.*

(2) ***Reflective split:*** *for all* `RoleSeparatedSkeleton` *inputs and H, the standard witness satisfies the autonomous split predicate* (Theorem 5.2)*.*

(3) ***Upstairs multiplicity (flagship):*** $\exists\, m_1 \neq m_2$ *with* $m_i.\mathsf{bridge.derivation}$ *equal in* $\mathcal{B}(A)$ (Theorem 5.3)*.*

(4) ***Strict refinement:*** $\exists\, x_1 \neq x_2 \in \mathcal{E}(A)$ *with* $\pi_A(x_1) = \pi_A(x_2)$ *on the NEMS spine; equivalently* $\text{ProperExtensionViaForgetful}(\pi_A)$ (Theorem 6.1)*.*

(5) ***Fiber geometry:*** $|\text{Fib}_A(b_A^*)| \geq 2$ *on NEMS; along* `splitInCanonicalFiber`*, role skeletons embed injectively* (Theorem 7.1, Theorem 7.2)*.*

(6) ***Generalization:*** *under* `RolePairDiverseCrownEligible` *and any H, the same non-injectivity conclusion holds; under* `RoleRichCrownEligible`*, nonempty enriched splits are obtained* (Theorem 8.2, Theorem 8.1)*.*

Reading the main cluster. Bare *certification* collapses: every standard route lands on the same canonical bare certificate. *Enriched realization* does not collapse: distinct enriched witnesses can share that bare image. The non-collapse is *measured* by non-injectivity of the typed forgetful map π_A and by nontriviality of the canonical fiber $\text{Fib}_A(b_A^*)$, with role data embedding injectively along a standard subfamily—not by informal "richness" talk.

In summary: the bare certification layer collapses to a unique canonical certificate; the enriched realization layer does not collapse—distinct routes produce genuinely distinct enriched witnesses; and the non-collapse is organized by the forgetful map π_A and its canonical fiber, where role data injects along a standard subfamily.

2.4 What is new here

The novelty is a **proved stratification**, not a type-theoretic platitude: collapse at the bare certification carrier (Theorem 4.1), strict refinement via π_A on enriched splits (Theorem 6.1, Theorem 8.2), and partial fiber classification along a standard subfamily (Theorem 7.2). Dependent types admit proof-relevant objects in principle; here the split/refinement/fiber facts are *theorems* about the library's certified carriers and the chosen comparison map—they are not imported as axioms of the framework. See Section 2.8 for contrast with barrier theorems and generic quotient talk; scope limits are Section 2.6.

2.5 Place in a broader series

Internal origin theorem. This paper establishes the internal architecture theorem—collapse at the bare carrier, strict refinement via the forgetful map, and fiber organization of multiplicity above the canonical base—*in the setting where that architecture is first proved as a load-bearing formal development*. It remains the flagship internal theorem paper; a reader-facing map of how the pieces assemble into one program is given in [7], with the sharpest summit statement of reflective non-exhaustion in [8].

Companion papers. Companion papers extend the same collapse/refinement/fiber analysis beyond the flagship Infinity Compression setting. An external validation paper [4] demonstrates that the architecture transfers cleanly across twelve independent mathematical families (quotients, products, bundles, coproducts, orbit quotients, pushouts, classifiers, localization, pullbacks). General-method papers [5, 6] then show that the architecture produces theorem-level results in group extensions (splitting iff trivial cocycle), embedding problems (solvable iff cocycle vanishes), and Quillen-style homotopy equivalence for Galois connections. Together the series moves from internal architecture theorem to external transfer to mathematical leverage on recognized problems.

2.6 What this paper does not claim (early scope)

We state scope early to avoid misreading. We do *not* classify all proof routes in the ambient logic; we do *not* prove surjectivity of role data onto the full fiber; we do *not* assert universal dominance over all formal frameworks; we do *not* claim the strongest possible reflective fixed-point closure. The theorems are conditional on the Infinity Compression architecture, the standard bundle, and the explicit crown-eligible / role hypotheses used in the Lean files. Section 12 and Section C expand this.

2.7 Why scope does not diminish the result

The significance of the paper does *not* require classifying every conceivable reflective framework. It lies in isolating and proving a reusable formal pattern—canonical collapse at one carrier, strict refinement via π_A at another, fiber-theoretic organization of multiplicity above a canonical base—together with: *(i)* a sharp structural distinction (certification vs. realization); *(ii)* the right carriers for the phenomenon; *(iii)* new comparison laws (forgetful map, non-injectivity, partial fiber classification); *(iv)* a layered theorem architecture that can be audited and reused; and *(v)* a witness-rich flagship

instance plus an honest generalized hypothesis class. That package is the contribution; broader universality is additional mathematics, not a prerequisite for the present theorems to matter.

2.8 Relation to incompleteness, fixed points, and certification

Classical metamathematics supplies *barrier* theorems (diagonalization, incompleteness): they exhibit limits or failures of closure. Fixed-point constructions yield existence of solutions to self-referential equations; certification in verification collapses many proofs to one output certificate. Those modes do not, by themselves, supply a *positive* classification of what remains above a *chosen* canonical certification carrier.

Contrast (three layers).

- **Generic proof relevance** (dependent type theory): richer types may carry nontrivial inhabitants in principle, without locating where distinctions vanish.
- **Proved collapse at one carrier:** Theorem 4.1 forces standard Phase-2 extraction to a *unique* bare certificate at $\mathcal{B}_{\text{can}}$—not a metaphor, a theorem on named types.
- **Proved non-injectivity at another carrier:** Theorem 6.1 / Theorem 8.2 show π_A is not injective on enriched splits; fibers organize the residue (Theorem 7.1, Theorem 7.2).

This paper's novelty is not "proof relevance exists" nor "some map is non-injective" in isolation. It is a **layered positive structure theorem**: (i) a *proved* canonical collapse at the bare certification carrier (Theorem 4.1); (ii) a *proved* strict refinement upstairs via a typed π_A and ProperExtensionViaForgetful (Theorem 6.1, Theorem 8.2); (iii) fiber nontriviality and *partial* fiber classification along a standard subfamily (Theorem 7.1, Theorem 7.2). The architecture is therefore: collapse downstairs, structured comparison and fibers upstairs—not a single barrier statement, and not merely informal quotient/fiber language.

Skeptical readers may still ask whether this is only "theorem vs. proof object" repackaged; the answer is that the paper fixes *named* carriers in a machine-checked library, proves collapse at $\mathcal{B}_{\text{can}}$, and then proves non-injectivity and fiber lemmas on *those* carriers with the same hypotheses used in the formal development (Section A).

Stratification in one sentence. The proved architecture is: *collapse* (where distinctions vanish at the bare record) $\to$ *split/refinement* (where they reappear in $\mathcal{E}(A)$ as non-injectivity of π_A) $\to$ *fiber organization* (where multiplicity sits in $\mathrm{Fib}_A(b_A^*)$ with injective role coordinates on a standard patch). That trichotomy is the paper's positive content; it is not the generic observation that DTT has proof relevance.

2.9 Main contributions (theorem-specific)

1. Phase-2 universal collapse to `admissibleStandard`; bundled CCP (Theorem 4.1, Theorem 4.3).
2. Reflective split from standard roles+extraction (Theorem 5.2).
3. Flagship upstairs multiplicity (Theorem 5.3); strict refinement / ProperExtensionViaForgetful (Theorem 6.1, Theorem 8.2).
4. Fiber nontriviality + role injection on standard subfamily (Theorem 7.1, Theorem 7.2).
5. Methodology: Section 9.

2.10 Roadmap

Section 3: notation $(\mathcal{E}, \mathcal{B}, \pi_A, b_A^*)$. Section 4–Section 7: collapse, split, π_A, fibers. Section 8: NEMS + generalization. Section 9: methods. Section 10: how the development instantiates its own distinction. Section 11: readings. Section 12–Section 14; the extended Section D closes the paper's program position. Appendices: Lean index, architecture sketch, audits, build, excerpts.

3 Framework and formal setting

3.1 Informal intuition

Fix a *compression architecture* A and admissible derivations for a principal invariant. Two aligned layers matter:

- **Certification:** a bare, theorem-level witness (proof-irrelevant at that carrier).
- **Realization:** proof-relevant data (roles, bridge, dependency shape) that may vary when the bare record is fixed.

Variation at the second layer is genuine iff it is not forced by the bare record alone; Section 4–Section 7 prove where each kind holds.

3.2 Formal notation aligned with the Lean carriers

Fix an architecture A in the sense of the Infinity Compression development. Write:

- $\mathcal{B}(A)$ for the type of bare certified records (in Lean, `AdmissibleSummitDerivation` together with the intended equivalence induced by proof irrelevance on its `Prop` fields).
- $\mathcal{E}(A)$ for the type of *enriched reflective splits*: witnesses `ReflectiveMirrorWitness` constrained by `ReflectiveSplitAutonomous`, packaged as `EnrichedReflectiveSplit` in the library.
- $b_A^* \in \mathcal{B}(A)$ for the distinguished library standard bare certificate (Lean: `admissibleStandard`), regarded as the canonical basepoint of the certification layer under the collapse theorems.
- $\pi_A : \mathcal{E}(A) \to \mathcal{B}_{\text{can}}(A)$ for the forgetful map `forgetToBareCanonical`, landing in the subtype $\mathcal{B}_{\text{can}}(A)$ of canonically certified bare records (`BareCanonicalCertificate`).

For $b \in \mathcal{B}_{\text{can}}(A)$, define the **fiber**

$$\text{Fib}_A(b) := \pi_A^{-1}(b) = \{x \in \mathcal{E}(A) : \pi_A(x) = b\}.$$

When $\mathcal{B}_{\text{can}}(A)$ collapses to a single point b_A^* for all standard constructions, the only fiber relevant for comparison is $\text{Fib}_A(b_A^*)$. Section 7 makes this explicit.

Forward pointer. The notation is architecture-agnostic. The flagship instance used throughout is a concrete NEMS-style spine with explicit `RoleSeparatedSkeleton` data (Section 8); the generalized laws replace that particular instantiation by crown hypotheses on A (Theorem 8.1).

3.3 Certification versus realization

Certification is uniqueness of the bare record at the theorem-level carrier: under the standard bundle and Phase-2 extraction interfaces, all admissible bare records coincide with b_A^*. **Realization** is proof-relevant data in $\mathcal{E}(A)$ not determined by $\pi_A(x)$ alone: non-injectivity of π_A is exactly the statement that realization content is not a function of bare certification.

3.4 Assumptions and scope

Results are proved in Lean 4 with dependent types. They concern a *precisely defined* class of structured reflective routes internal to the Infinity Compression definitions (architectures, admissible derivations, crown-eligibility hypotheses, role-separated skeletons, extraction interfaces). That restriction is a strength: collapse, comparison map, and fiber analysis are proved on explicitly represented carriers, not left at the level of informal "proof" metaphors. We do not pretend to quantify over arbitrary proof trees of pure logic; we quantify over the library's typed routes—where the theorems live.

Global arc. Sections 4–7 prove, in order: collapse at the bare carrier; split on enriched witnesses; strict refinement (non-injectivity of π_A); then fiber-theoretic organization of the resulting multiplicity (nontrivial canonical fiber, injective roles on a standard subfamily).

4 Canonical certification

4.1 Canonical certification theorem (universal collapse at the bare carrier)

Let A be a compression architecture. The Phase-2 construction `admissibleFromSkeleton` takes a constructive skeleton s and an extraction interface H (supplying the two named shard witnesses) and produces a bare admissible record in $\mathcal{B}(A)$.

Theorem 4.1 (Canonical certification / universal collapse)**.** *For every constructive skeleton s and every `SummitComponentExtraction` H,*

$$\text{admissibleFromSkeleton}(s, H) = b_A^*,$$

the library standard bare record `admissibleStandard`. In particular, the bare record is independent of s along this interface.

Lean: `admissibleFrom_skeleton_eq_admissibleStandard` (`MetaProof/LocalToGlobalDerivation.lean`); proof is `simp` after unfolding definitions—the mathematical content is that the two `Prop` fields are fixed targets, so proof irrelevance collapses all variation at `AdmissibleSummitDerivation`. See Section B.

4.2 Collapse as a structural law, not a bug

Here collapse is stratification, not failure: the bare carrier is a quotient-level invariant; irreducibility is tracked on proof-relevant types above it (Section 9.4).

4.3 Canonical certification principle (bundled interfaces)

Definition 4.2 (Canonically certified bare record)**.** Write $b_A^* :=$ `admissibleStandard` for the library standard packaged record. Then $\text{Can}(d)$ means $d = b_A^*$; equivalently `IsCanonicalBareSummitCertification` d (definitionally $d =$ `admissibleStandard` in `CanonicalCertification.lean`).

Theorem 4.3 (Phase-2 canonical certification principle)**.** *For fixed A, every `admissibleFromSkeleton` path and every roles+extraction bridge derivation lands in* Can*:*

$$\forall s, H.\ \text{Can}(\text{admissibleFromSkeleton}(s, H)),$$

$$\forall \mathsf{roles}, H.\ \text{Can}(\mathsf{SkeletonIndexedComponentWitness.fromRolesExtraction}(\mathsf{roles}, H).\mathsf{derivation}).$$

Here `SkeletonIndexedComponentWitness.fromRolesExtraction` is the library's packaged bridge from `RoleSeparatedSkeleton` + `SummitComponentExtraction` to a skeleton-indexed witness whose `derivation` field is the bare record compared in collapse theorems (`CanonicalCertification.lean`, `SkeletonIndexedExtractionBridge.lean`).
Lean: `canonical_certification_principle_phase2` (`MetaProof/CanonicalCertification.lean`).

4.4 Consequence

Irreducible comparison problems for reflective structure cannot be stated as distinctions between bare certificates in this standard-bundle regime. They lift to $\mathcal{E}(A)$ and to $\mathrm{Fib}_A(b_A^*)$.

5 Enriched reflective realization

5.1 Enriched reflective witnesses

An *enriched reflective realization* is a proof-relevant witness (Lean: `ReflectiveMirrorWitness`) packaging a bridge derivation in $\mathcal{B}(A)$ together with role-separated skeleton data, dependency-shape recording, and autonomous-mirror constraints. It is not a mere annotation of a bare record: inequality of witnesses can occur at fixed bare derivation (Theorem 5.3).

5.2 Reflective split theorem

Definition 5.1 (Reflective split (autonomous))**.** A witness m *splits reflectively* when its bridge derivation is canonically certified and m is an autonomous summit mirror (Lean: `ReflectiveSplitAutonomous`).

Theorem 5.2 (Standard roles + extraction yield split)**.** *For every* `RoleSeparatedSkeleton` ρ *and every* `SummitComponentExtraction` H*,*

$$\mathtt{ReflectiveSplitAutonomous}\big(\mathtt{ReflectiveMirrorWitness.fromRolesAndExtraction_standard}(\rho, H)\big).$$

Lean: `reflective_split_from_roles_standard` (`MetaProof/ReflectiveSplit.lean`).

5.3 Upstairs irreducibility (flagship)

Upstairs irreducibility means existence of $m_1 \neq m_2$ with the same bridge derivation in $\mathcal{B}(A)$.

Theorem 5.3 (Distinct enriched realizations, identical bare bridge)**.** *On the NEMS flagship spine, for suitable extraction H, there exist standard witnesses $m_1 \neq m_2$ with m_1.bridge.derivation $=$ m_2.bridge.derivation.*

Lean: `nems_distinct_autonomous_splits_same_canonical_bare` (`MetaProof/EnrichedIrreducibility.lean`, **T-C+.5**); proof composes `reflective_split_from_roles_standard` with bridge equality from `skeleton_indexed_fromRoles_derivation_eq_admissibleStandard` and witness inequality from `reflective_mirror_from_roles_ne_of_roles_ne` + `nems_role_skeletons_ne`.

5.4 Corollary (theorem-derived)

Corollary 5.4 (Canonicality of certification does not imply uniqueness of realization)**.** *Bare derivation equality does not force witness equality. Theorem 5.3 is a formal witness.*

6 Strict refinement of enriched reflective routes

6.1 Typed route spaces

Work with subtypes for clean equality: $\mathcal{B}_{\text{can}}(A)$ is implemented as `BareCanonicalCertificate` (bare derivation + `IsCanonicalBareSummitCertification`), and $\mathcal{E}(A)$ as `EnrichedReflectiveSplit` (witness + `ReflectiveSplitAutonomous`). This keeps the comparison map π_A a genuine function on typed carriers rather than a loose informal "forgetting."

6.2 The forgetful map as organizing object

Define $\pi_A : \mathcal{E}(A) \to \mathcal{B}_{\text{can}}(A)$ by

$$\pi_A(x) := \big(x.\mathsf{val.bridge.derivation},\ \text{witness of canonicity}\big).$$

Soundness is layered: **no-loss to canonicity** (`reflective_split_forgets_to_canonical`, **T-DQ4.1**); **alignment of the forgetful projection with** `bridge.derivation` (`forgetToBareCanonical_sound`, **T-DQ4.2**); and equality of forgetful images from equal bridge derivations (`forgetToBareCanonical_eq_of_bridge_derivation_eq`). On the standard development, enriched splits built from roles+extraction forget to `canonicalBareCertificate` (`enriched_reflective_split_forgets_to_canonical_bare_certificate`, `from_roles_standard_forgets_to_canonical_bare_certificate`).

6.3 Strict refinement (non-injectivity)

Theorem 6.1 (Strict refinement / proper extension on NEMS)**.** *There exist* $x_1 \neq x_2 \in \mathcal{E}(A_{\text{NEMS}})$ *with* $\pi_{A_{\text{NEMS}}}(x_1) = \pi_{A_{\text{NEMS}}}(x_2)$.

Lean: `nems_reflective_route_not_injective_to_bare` and `nems_reflective_route_strictly_refines_bare`, the latter repackaging the former as `ProperExtensionViaForgetful` ("$\exists x_1 \neq x_2,\ \pi(x_1) = \pi(x_2)$").

6.4 Why non-injectivity is the right strictness notion

Injective π_A would mean enriched data is (recoverably) a function of bare certification; then the enriched layer would be redundant up to encoding. Non-injectivity is the precise statement that *more than one* enriched solution exists over the same certified base—a failure of uniqueness for realization even when certification is unique. This matches the fiber viewpoint: fibers have cardinality > 1 iff π_A is not injective over the relevant preimage problem.

6.5 Answer to the "just tags" objection

Theorem 5.3 and Theorem 6.1 refute the "mere tags" reading: distinct witnesses at equal bare image.

7 Fiber structure and partial classification

7.1 Basepoint and preimage

Fix $b \in \mathcal{B}_{\mathrm{can}}(A)$. The *fiber* of π_A at b is $\mathrm{Fib}_A(b) = \pi_A^{-1}(b)$, implemented as a subtype (`fiberAtBare`). Program C+ collapse implies that for standard enriched routes, $\pi_A(x)$ is the distinguished basepoint b_A^* (`canonicalBareCertificate`); thus the induced classification problem is not "all fibers in sight" but the *canonical* fiber $\mathrm{Fib}_A(b_A^*)$.

7.2 Nontriviality

Theorem 7.1 (Nontrivial canonical fiber (NEMS))**.** *For suitable H, the set* $\mathrm{Fib}_{A_{\mathrm{NEMS}}}(b^*_{A_{\mathrm{NEMS}}})$ *contains at least two distinct points.*

Lean: `nems_fiber_over_canonical_nontrivial` (`MetaProof/ReflectiveFiberClassification.lean`), repackaging non-injectivity of π as inequality of fiber points.

7.3 What is (and is not) classified

We do *not* classify all of $\mathrm{Fib}_A(b_A^*)$, establish surjectivity of roles onto the fiber, or list all enriched splits. What we prove is a **role-sensitive partial classification** on the standard subfamily "roles + extraction + split hypothesis": embedding roles into the fiber along `splitInCanonicalFiber` is injective.

Theorem 7.2 (Role injection into canonical fiber (general + NEMS corollary))**.** *Let H be fixed; let* $\mathsf{splitInCanonicalFiber}(H, \mathsf{roles}, \mathsf{hs})$ *denote the standard fiber point from roles + split data. For all* $\mathsf{roles}_1, \mathsf{roles}_2$ *and admissible split hypotheses* $\mathsf{hs}_1, \mathsf{hs}_2$*,*

$$\mathsf{splitInCanonicalFiber}(H, \mathsf{roles}_1, \mathsf{hs}_1) = \mathsf{splitInCanonicalFiber}(H, \mathsf{roles}_2, \mathsf{hs}_2) \implies \mathsf{roles}_1 = \mathsf{roles}_2.$$

Lean: `splitInCanonicalFiber_roles_injective` (**T-FC.4**); NEMS corollary `nems_roles_injective_into_canonical_fiber`. Distinct roles yield distinct fiber points via `nems_distinct_roles_distinct_fiber_points` (**T-FC.2**).

7.4 Consequence

Along the standard embedding, roles coordinate distinct fiber points; full enumeration of $\mathrm{Fib}_A(b_A^*)$ remains open (Section 12).

8 Flagship architecture and structural generalization

8.1 Canonical flagship instance (NEMS spine)

The **NEMS spine** is used not as external authority but as a *witness-rich*, fully formalized architecture: explicit `RoleSeparatedSkeleton` data, crown-eligible hypotheses, dual poles, and bridge lemmas feeding Program C+. Its role is analogous to a canonical model in algebra or logic: it shows the abstract structural law is *non-vacuous* and supports the strongest concrete witness theorems in the paper (Theorem 5.3, Theorem 6.1, Theorem 7.1) under explicit extraction hypotheses. It is not assumed universal; it is the first inhabited model in which the phenomenon is exhibited with full skeletal and bridge detail.

8.2 Why the flagship matters

A comparison law with no inhabited instance would risk vacuity. NEMS forces the split/refinement/-fiber pattern in a completely specified development—not a hypothetical sketch.

8.3 Generalized classes: honest intermediates

Definition 8.1 (Role-rich / role-pair-diverse crown eligibility)**.** *Role-rich crown eligibility* (`RoleRichCrownEligible`) is `CrownEligible` together with `Nonempty` (`RoleSeparatedSkeleton` A). *Role-pair-diverse crown eligibility* (`RolePairDiverseCrownEligible`) is `CrownEligible` together with *two* distinct `RoleSeparatedSkeleton` values—the minimal explicit "two upstairs coordinates" hypothesis for generic strict refinement (`GeneralizedReflectiveFiberRefinement.lean`).

Theorem 8.2 (Generalized strict refinement)**.** *If A satisfies* `RolePairDiverseCrownEligible` *and H :* `SummitComponentExtraction` *A, then*

$$\exists\, x_1 \neq x_2 \in \mathcal{E}(A), \quad \pi_A(x_1) = \pi_A(x_2),$$

and hence `ProperExtensionViaForgetful` *holds for π_A.*

Lean: `role_pair_diverse_crown_eligible_strict_refinement` (**T-GR.3**), `role_pair_diverse_crown_eligible_proper_extension_forget` (**T-GR.4**, `ProperExtensionViaForgetful`); NEMS instances `nems_role_pair_diverse_strict_refinement`, `nems_spine_role_pair_diverse_crown_eligible` (**T-GR.7**). Role-rich: `role_rich_crown_eligible_obtains_enriched_reflective_split` (**T-GR.1**), `role_rich_crown_eligible_nonempty_enriched_split`. Generalized fiber injection alias: `generalized_role_injection_into_canonical_fiber` (**T-GR.5**, restates **T-FC.4**).

8.4 Why not stop at `CrownEligible` alone?

`CrownEligible` alone does not assert two distinct `RoleSeparatedSkeleton` values. Theorem 8.2 needs a pair (r_1, r_2) with $r_1 \neq r_2$ so that `reflective_mirror_from_roles_ne_of_roles_ne` can separate witnesses while `skeleton_indexed_fromRoles_derivation_eq_admissibleStandard` collapses the bare bridge—that is exactly `RolePairDiverseCrownEligible`. This is weaker than classifying $\mathrm{Fib}_A(b_A^*)$.

8.5 Concrete and abstract together

The generalized theorems are not "more fundamental" than the flagship: Theorem 8.2 isolates the *structural conditions* (crown + two distinct role skeletons + extraction) under which the same non-injectivity pattern necessarily appears. The NEMS spine is the canonical *witness-rich* instance: fully named architectures, explicit skeletons, and the strongest concrete equalities in Theorem 5.3–Theorem 7.1. Those flagship results are not mere corollaries of a weaker abstraction—they are the reference model; `nems_spine_role_pair_diverse_crown_eligible` shows NEMS satisfies the hypotheses of Theorem 8.2, linking abstract law to inhabited structure.

9 Architectural and methodological contributions

Each item below pairs a *theorem pattern* (what is proved) with a *methodological lesson* (what to reuse elsewhere).

9.1 Layered proof architecture (theorem-driven)

Theorem pattern: a stack of carriers from `Frontier/` invariants through Phase-2 collapse Theorem 4.1 to Program C+ split/refinement/fiber. **Lesson:** state the audit question "does this carrier retain the distinction?" at each layer before moving up.

9.2 Composable polarities (load-bearing)

Theorem pattern: `autonomous_reflective_split_records_non_monopolar_standard` (**T-C+.4**) + `RolePairDiverseCrownEligible` feeding Theorem 8.2. **Lesson:** link semantic poles in dependency shape to distinct role skeletons, then use bare-bridge collapse to force equal π_A-images.

9.3 Carrier sensitivity

Theorem pattern: collapse at $\mathcal{B}(A)$ (Theorem 4.1) vs. multiplicity in $\mathcal{E}(A)$ (Theorem 5.3) and $\mathrm{Fib}_A(b_A^*)$ (Theorem 7.1). **Lesson:** *survival of distinctions* is itself a theorem—prove where they vanish and where they persist; do not conflate informal "same route" across types.

9.4 Collapse audit as a formal method

Theorem pattern: propositions of the form "D vanishes at carrier C"; instance Theorem 4.1 with $C =$ `AdmissibleSummitDerivation`. **Lesson:** collapse audits are negative theorems that *license* relocation of D to $\mathcal{E}(A)$ or fibers—exported formal method, not commentary.

9.5 Route comparison methodology

Theorem pattern: typed π_A, soundness lemmas, then ProperExtensionViaForgetful via Theorem 6.1 / Theorem 8.2. **Lesson:** formal information loss / refinement is $\neg\mathrm{Inj}(\pi_A)$ on chosen routes—prove it as a comparison theorem, not as a metaphor for "less detail."

10 Reflexive self-location of the theorem program

The paper proves that canonical certification at the bare carrier does not exhaust reflective structure: above a canonically certified base there remains a realization space that must be analyzed through route comparison and fibers (Theorem 2.1). The *formal development itself* exhibits the same pattern. The final theorem-level statements recorded at the bare certification carrier are not the whole mathematical content of the theory; the load-bearing mathematical structure lies in the layered route through collapse, split, refinement, and fiber analysis—the very layers the paper formalizes. In this precise sense, the theorem program **instantiates the distinction it proves**: certification at one stratum, irreducible structure above it, accessed only through enriched routes and comparison geometry.

Once collapse at $\mathcal{B}(A)$ is established (Theorem 4.1), a mathematically serious continuation that asks reflective comparison questions in this framework is *structurally directed* toward enriched carriers, typed π_A, and fiber-theoretic organization—not because of stylistic preference, but because collapse *locates* where distinctions cannot live. The present development follows that route. This is **not** a claim that the development is the unique possible proof architecture (no uniqueness theorem for "all of mathematics" is offered), nor that the library "explains itself" in any metaphysical sense. It *is* the claim that the program's own proof architecture is an example of the certification/realization

split the theory establishes, and that the move upstairs is the natural forced continuation once bare collapse is a theorem.

Bridge. The point is not that "one carrier is richer" in the abstract; it is that the library proves *exactly where* distinctions vanish (bare certification) and *exactly where* they survive (enriched routes and fibers), on explicit hypotheses and machine-checked carriers.

11 Significance across fields

Each paragraph: *theorem claim* → one *reading*.

11.1 Logic and metatheory

Theorem 4.1–Theorem 7.1 realize a quotient/fiber pattern for reflective meta-theory without a new incompleteness theorem: certificate layer vs. fibered multiplicity above b_A^*.

11.2 Computer science and formal methods

Theorem 6.1: $\neg\mathrm{Inj}(\pi_A)$ for the certification forgetful map—forgetful APIs can identify distinct implementations.

11.3 Multi-agent verification and layered audits

Theorem 6.1 + Theorem 7.2: one aggregated bare verdict, multiple structured realizations; roles inject on a standard patch.

11.4 Foundations

Theorem 5.4, Theorem 6.1: *canonicality without flattening*—canonical base, non-unique enriched realization.

12 Limitations and open directions

12.1 Not claimed

This paper proves a *strong structural law* for a precise formal class: the carriers and hypotheses cited in Section A. Stronger universal statements—classifying all of $\mathrm{Fib}_A(b_A^*)$, surjectivity of roles onto the full fiber, maximal fixed-point strength, comparison across arbitrary formalisms—are left open not because the present package is weak, but because they are *different* mathematical targets. The load-bearing theorems are conditional; within that scope they are sharp.

In particular, we do *not* assert that the standard bare certificate fails for *every* conceivable question in other frameworks. We assert that, for the Infinity Compression standard bundle and the routes formalized here, the nontrivial comparison geometry for the phenomena we study lives on enriched carriers and fibers (Theorem 2.1). That is a positive, scoped theorem—not an apology.

12.2 Packaging-sensitive mirror caveat

As recorded in the library epistemology (see `IC_Closed_Open_Ledger.md` in the repository notes), some mirror/crown lemmas still use the library standard derivation as a witness interface. This does *not* undermine the Program C+ split/refinement/fiber theorems, but it does bound what can be

claimed about *independence* of mirror routes from the standard summit lemma package. The paper avoids over-claiming that independence.

12.3 Fiber classification frontier

Exact cardinality / structure of $\mathrm{Fib}_A(b_A^*)$ beyond the injective standard subfamily is open. Sharpening this is the natural next mathematical step after nontriviality + partial classification.

12.4 Natural next directions

Stronger comparison laws (dominance between reflective architectures), carrier redesign with audited distinctions, and broader synthesis with external certification frameworks are left to future work.

13 Discussion: auditability vs. slogans

Collapse is falsifiable. Theorem 4.1 refutes skeleton-parameter sensitivity at the bare record; without such a theorem, "collapse" would be rhetoric.

Strict refinement is a comparison theorem. Theorem 6.1 is non-injectivity of typed π_A, not informal "richness."

Fibers locate multiplicity. Theorem 7.1 and Theorem 7.2 place multiplicity in $\mathrm{Fib}_A(b_A^*)$ with partial coordinates, not in generic prose.

14 Summary

This paper's local result, in one line of audit language: Phase-2 collapse at the bare carrier; strict refinement of π_A on enriched splits; nontrivial canonical fiber with partial role injection along a standard patch (Theorem 2.1). After collapse, comparison belongs upstairs in $\mathcal{E}(A)$ and in fibers—not in ad hoc bare-field distinctions. The development instantiates the split it proves (Section 10).

A Theorem and definition index (paper ↔ Lean)

Paper ref.	Informal statement	Lean name	Module (under `InfinityCompression/`)
Theorem 4.1	Phase-2 skeleton path equals `admissibleStandard`	`admissibleFrom_skeleton_eq_admissibleStandard` (**T-P3.7**)	`MetaProof/LocalToGlobalDerivation.lean`
—	Corollary: skeleton-irrelevant equality	`admissibleFrom_skeleton_eq_irrespective_of_skeleton`	same
Theorem 4.3	Bundled Phase-2 + roles bridge canonicality	`canonical_certification_principle_phase2` (**T-C+.1**)	`MetaProof/CanonicalCertification.lean`
—	Bridge derivation canonical	`fromRoles_extraction_bridge_derivation_canonical`	same
Theorem 4.2	"Bare equals standard" predicate	`IsCanonicalBareSummitCertification` (**D-C+.1**)	same
Theorem 5.2	Standard roles+extraction ⇒ split	`reflective_split_from_roles_standard` (**T-C+.3**)	`MetaProof/ReflectiveSplit.lean`
—	Split predicate	`ReflectiveSplitAutonomous`	same

Paper ref.	Informal statement	Lean name	Module
—	Witness type	`ReflectiveMirrorWitness`, `fromRolesAndExtraction_standard`	`MetaProof/ReflectiveMirrorWitness.lean`
Theorem 5.3	Two splits, same bare derivation, witnesses $\neq$	`nems_distinct_autonomous_splits_same_canonical_bare` (**T-C+.5**)	`MetaProof/EnrichedIrreducibility.lean`
—	Non-monopolarity on standard shape	`autonomous_reflective_split_records_non_monopolar_standard` (**T-C+.4**)	same
Route comparison (EPIC_ 008 / DQ4)			
—	Typed bare/enriched routes	`BareCanonicalCertificate`, `EnrichedReflectiveSplit` (**D-DQ4.1–2**)	`MetaProof/ReflectiveRouteComparison.lean`
—	Proper extension predicate	`ProperExtensionViaForgetful` (**D-DQ4.4**)	same
—	Forgetful map π_A	`forgetToBareCanonical` (**D-DQ4.3**)	same
—	No-loss + soundness	`reflective_split_forgets_to_canonical`, `forgetToBareCanonical_sound`	same
Theorem 6.1	Non-injectivity / strict refinement NEMS	`nems_reflective_route_not_injective_to_bare`, `nems_reflective_route_strictly_refines_bare`	same
Fiber classification (EPIC_ 009 / FC8)			
Theorem 7.1	$\lvert\mathrm{Fib}(b^*)\rvert \geq 2$ on NEMS	`nems_fiber_over_canonical_nontrivial` (**T-FC.3**)	`MetaProof/ReflectiveFiberClassification.lean`
—	Forget to canonical bare certificate	`enriched_reflective_split_forgets_to_canonical_bare_certificate` (**T-FC.1a**)	same
—	Distinct roles $\Rightarrow$ distinct fiber points	`nems_distinct_roles_distinct_fiber_points` (**T-FC.2**)	same
Theorem 7.2	Roles injective into fiber embedding	`splitInCanonicalFiber_roles_injective` (**T-FC.4**)	same
—	NEMS corollary	`nems_roles_injective_into_canonical_fiber`	same
—	Fiber / basepoint	`fiberAtBare`, `canonicalBareCertificate`, `splitInCanonicalFiber`	same
Generalized refinement (EPIC_ 010 / GR2)			
Theorem 8.2	Pair-diverse $\Rightarrow$ strict refinement	`role_pair_diverse_crown_eligible_strict_refinement` (**T-GR.3**)	`MetaProof/GeneralizedReflectiveFiberRefinement.lean`
—	Pair-diverse $\Rightarrow$ `ProperExtensionViaForgetful`	`role_pair_diverse_crown_eligible_proper_extension_forget` (**T-GR.4**)	same
Theorem 8.1	Intermediate hypotheses	`RoleRichCrownEligible`, `RolePairDiverseCrownEligible`	same
—	NEMS satisfies hypotheses	`nems_spine_role_rich_crown_eligible`, `nems_spine_role_pair_diverse_crown_eligible` (**T-GR.7**)	same
—	Role-rich $\Rightarrow$ nonempty splits	`role_rich_crown_eligible_obtains_enriched_reflective_split` (**T-GR.1**)	same
—	Generalized injection alias	`generalized_role_injection_into_canonical_fiber` (**T-GR.5**)	same
Bridge collapse used throughout			
—	Roles+extraction derivation collapse	`skeleton_indexed_fromRoles_derivation_eq_admissibleStandard` (**T-P3.5c**)	`MetaProof/SkeletonIndexedExtractionBridge.lean`

B Broader Infinity Compression architecture (sketch)

The verified Infinity Compression library spans landscape/summit statements, route-B meta theory, crown and mirror constructions, Program E/006E infrastructure, and the Program C+ / route-comparison / fiber layers. **This paper's mathematical contribution is concentrated in the latter block:** canonical certification vs. enriched realization, strict refinement of π_A, and partial fiber classification—together with the collapse audits that justify moving comparison upstairs. The list below situates that block inside the larger stack for readers of the repository; it is not required for reading the main theorems, which are self-contained modulo the cited `MetaProof` modules.

Layers are *logical* (not a runtime stack):

- **Summit / landscape (`Frontier/`):** principal certified invariant statements (e.g. `ic_unive`

rsal_theorem_summit, ICUniversalTheorem.lean); landscape aggregation; anti-translation abstracts.

- **Route B meta (MetaProof/):** summit bundles (SummitTargets), dependency shape (DependencyShape), admissible records (AdmissibleDerivations), route-B first theorems (RouteBFirstTheorems). Feeds Phase-2 extraction.
- **Crown / derivation necessity:** CrownEligible, ArchitectureDerivationCorrespondence, DerivationNecessity, DerivationInevitability, MetaCrown.
- **Program E / 006E (supporting):** ProgramECrownSummit, ProgramE006EReflectiveFixedPoint—summit-capable mirrors and reflective fixed-point packaging; cited in the paper as *infrastructure*, not as the main headline.
- **Post-summit autonomous mirror:** ReflectiveMirrorWitness, AutonomousMirrorConstruction; standard bridge lemmas (reflective_mirror_bridge_derivation_eq_admissibleStandard).
- **Program C+ (this paper's core):** CanonicalCertification, ReflectiveSplit, EnrichedIrreducibility, SkeletonIndexedExtractionBridge, LocalToGlobalDerivation.
- **Route comparison (EPIC_008):** ReflectiveRouteComparison—forgetToBareCanonical, ProperExtensionViaForgetful.
- **Fiber classification (EPIC_009):** ReflectiveFiberClassification—fiberAtBare, splitInCanonicalFiber, role injection.
- **Generalized refinement (EPIC_010):** GeneralizedReflectiveFiberRefinement—RoleRichCrownEligible, RolePairDiverseCrownEligible and generic strict refinement.

C Collapse audits, falsified hypotheses, and open packaging

Carrier / layer	Rigidity (what collapses)	Where comparison structure lives
AdmissibleSummitDerivation (Phase-2 + standard bundle)	Uniquely admissibleStandard (Theorem 4.1)	No nontrivial bare certificate comparison
ReflectiveMirrorWitness / EnrichedReflectiveSplit	Bridge derivations still canonical	Distinct witnesses, same bare image (Theorem 5.3, Theorem 6.1)
fiberAtBare / splitInCanonicalFiber	Basepoint b_A^* fixed for standard routes	Nontrivial fiber; injective roles on standard subfamily (Theorem 7.1, Theorem 7.2)

Table 2: Carrier sensitivity at a glance: collapse downstairs vs. structured comparison upstairs.

What collapsed (proved): For every ConstructiveDerivationSkeleton s and SummitComponentExtraction H, admissibleFromSkeletonsH=admissibleStandard (admissibleFrom_skeleton_eq_admissibleStandard, Theorem 4.1). For roles+extraction bridges, skeleton_indexed_fromRoles_derivation_eq_admissibleStandard forces the same bare record. The bundled statement is canonical_certification_principle_phase2 (Theorem 4.3).

What was falsified (as a design hypothesis): the expectation that one could read off nontrivial "in-record" distinctions from skeleton parameters *at* the type `AdmissibleSummitDerivation` in the standard-bundle regime; the two `Prop` shard fields are fixed targets, so proof irrelevance collapses the record.

Correct carrier for multiplicity: `ReflectiveMirrorWitness` / `EnrichedReflectiveSplit`, fibers `fiberAtBare` over `canonicalBareCertificate`, and role-sensitive maps `splitInCanonicalFiber`.

Why enriched level is the right carrier: `IsCanonicalBareSummitCertification` is definitionally `d=admissibleStandard`; all standard Phase-2 routes hit that predicate. Distinctions that matter for comparison geometry are therefore tracked on proof-relevant types above that quotient.

Packaging mirror caveat (audit): as documented in `specs/NOTES/IC_Closed_Open_Ledger.md` (repository ledger note), several crown/mirror lemmas still use the library standard derivation as a witness interface. This does *not* undermine the Program C+ split/refinement/fiber theorems proved here, but it bounds claims about *independence* of mirror routes from the principal summit lemma package. The paper states this in Section 12.2.

What remains open: full classification of $\mathrm{Fib}_A(b_A^*)$; surjectivity of role data onto the fiber; stronger mirror-route independence; architecture-dominance comparison laws beyond the hypotheses in `ReflectiveRouteComparison` / `GeneralizedReflectiveFiberRefinement`.

D Conclusion

Role in the full program. This flagship manuscript is where the architecture is first proved as a load-bearing internal development (not the suite map). The *transferability paper* [4], *general-method manuscripts* [5, 6], and the *synthesis* / *summit* manuscripts [7, 8] extend and contextualize that origin without replacing it. All main results here are machine-checked in Lean 4 with zero `sorry` in proof terms under `InfinityCompression/` on explicit hypotheses.

A Artifact and build details

- **Repository layout:** the Lean project lives at `infinity-compression/infinity-compression-lean/` relative to the `infinity-compression` monorepo root. This paper's `.tex` file is under `infinity-compression-lean/papers/Canonical_Certification_and_Enriched_Reflective_Realization/`.
- **Root module:** `InfinityCompression.lean` (import closure of the library).
- **Lean toolchain (pinned):** `leanprover/lean4:v4.29.0-rc6` (`infinity-compression-lean/lean-toolchain`).
- **Mathlib pin:** `require mathlib` at tag `v4.29.0-rc6` in `lakefile.lean` (same tag family as the Lean toolchain).
- **Build command:** from `infinity-compression-lean/`, run `lakebuild`

- `sorry` **policy (verified):** `README.md` states zero `sorry` in proof terms under `InfinityCompression/`; the Program C+ slice referenced in this paper is part of that scope. Axioms are standard Mathlib + Lean kernel only (no custom axioms for these results).
- **Epistemology / packaging audit:** `specs/NOTES/IC_Closed_Open_Ledger.md` (mirror packaging, closed vs. open claims).
- **Theorem inventory cross-check:** `specs/NOTES/EPIC_THEOREM_INVENTORY_LEAN_PAPER_PREP.md` (spine names and module paths).
- **Primary modules for this paper:** `InfinityCompression/MetaProof/` modules (`*.lean`) as listed in Section A.

B Selected Lean excerpts

```
theorem admissibleFrom_skeleton_eq_admissibleStandard {BD : Type u} {n :
   Nat}
    {A : CompressionArchitecture BD n} (s : ConstructiveDerivationSkeleton
        A)
    (H : SummitComponentExtraction A) :
    admissibleFromSkeleton s H = admissibleStandard := by
  simp [admissibleFromSkeleton, admissibleStandard]
```

```
def forgetToBareCanonical {BD : Type u} {n : Nat} {A :
   CompressionArchitecture BD n}
    (x : EnrichedReflectiveSplit BD n A) : BareCanonicalCertificate :=
  Subtype.mk x.val.bridge.derivation x.property.left
```

In the library source this is written with the anonymous constructor `<...>` for `Subtype` / pair syntax; `x.property.left` is the `IsCanonicalBareSummitCertification` component of `ReflectiveSplitAutonomous` (`ReflectiveRouteComparison.lean`). The definitional expansion matches `<...>` after `Subtype` $\leftrightarrow$ pair unpacking.

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